# OLIVIA WANG

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## EDUCATION

**California Institute of Technology** *Pasadena, CA* | GPA: 4.0/4.0 **Jun 2027**

*B.S. in Computer Science*

- Courses: Deep Learning, Diffusion, Operating Systems, Algorithms, Probabiliy, Law & Finance for Startups

## EXPERIENCE

**Software Development Intern | Amazon Music** *Culver City, CA* **Jun 2026 - Present**

- Engineered an end-to-end mobile playback feature using React Native and a distributed, service-oriented architecture, delivering seamless on-demand playlist and track data to millions of active listeners.

**Machine Learning Researcher | Perona Vision Lab, Caltech** *Pasadena, CA* **Jan 2025 - Present**

- Architected PyTorch-based infrastructure to benchmark vision-language diffusion, autoregressive models against human reasoning datasets across 10+ visual tasks.
- Developed a high-precision human-study application for 100 users using Python & Javascript for web-based data collection with precise stimuli timing.
- Engineered Python backends to optimized GPU system performance for batch processing, distributed data parallelization, and concurrent execution on HPC.

**Software Engineering Intern | Eli Lilly's Protomer Technologies** *Pasadena, CA* **Jan 2025 - Jun 2026**

- Architected and deployed a full-stack web application using Python (Dash) to visualize high-frequency glucose sensor data, enabling hardware-in-the-loop experiments for critical medical R&D.
- Built a multi-threaded backend in Python to ingest BLE sensor data and deliver real-time ML model predictions for automated insulin regulation logic.

## PUBLICATIONS

**Wang, O.**, et al. "Do Vision Language Models see like us? Yes, if we look carefully." In Review.
**Wang, O.**, Larsson, E., Murray, R. "A DNA Part Library for *X. griffiniae*." *ACS Synthetic Biology. (2025)*.

## PROJECTS

**Spectrally Consistent Cloud Removal via NAMM-EDM** | *PyTorch, SLURM, CUDA* Apr 2026 - Jun 2026

- Achieve state of art metrics SAM of 5.008 and PSNR of 32.43 dB across 13 competing methods with Neural Approximate Mirror Map integrated with EDM diffusion model to enforce spectral consistency.
- Engineered a 3-phase training pipeline on HPC with a custom constraint loss to preserve physical consistency in generated Sentinel-2 imagery.

**Sheet Engine with Automated Logic Validation** | *Python, Pytest, Unittest, API* Jan 2026 - March 2026

- Led a 3-person team to engineer a spreadsheet calculation engine with complex formula parsing and dependency tracking, optimizing for speed and scale.
- Developed automated unit & integrated testing suite, ensuring 100% calculation accuracy and system reliability across edge cases.

## SKILLS

- **Software:** Python, Java, C, Javascript, Git, Linux, Containerization, React, Automated Tesing
- **Machine Learning:** PyTorch, Tensorflow, scikit-learn, Deep Learning, Computer Vision, LLM, NLP, AWS

## LEADERSHIP

- Machine Learning Teaching Assistant for 100+ students, Caltech Jan 2026 - Present
- Health Advocate, Social Media & Admissions Ambassador, Caltech May 2024 - Present
- Team USA and National Figure Skater, Los Angeles Ice Theater Sept 2018 - Jun 2024